

# Valuation of Options - part 3

## Quantitative Finance

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# Assumptions Black-Scholes market (recap)

In class we will almost exclusively work with the 'Black-Scholes market'.

## Assumptions on price processes:

Asset price:

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad \text{var}(W_1) = 1$$

Money Market Account:

$$dB_t = rB_t dt, \quad B_0 = 1$$

## Assumptions on market:

- frictionless trading
  - no transaction costs
  - trading in continuous-time is possible
  - no restrictions on short sales and fractional positions
- no credit/counterparty risk: borrowing rate = lending rate

- we are interested in the price  $C_t$ , for  $t \in [0, T)$ , of (European) option with payoff  $h(S_T)$  at expiration date/maturity  $T$
- we will discuss three methods to determine fair price (using continuous-time model for financial market):
  - Black-Scholes Partial Differential Equation
  - risk-neutral pricing
  - pricing kernel

**These slides discuss risk neutral pricing and the pricing kernel approach.**

The logic of risk-neutral pricing:

- ➊ **First Fundamental Theorem:** no arbitrage  $\Leftrightarrow$  there is a measure  $\mathbb{Q} \sim \mathbb{P}$  under which *discounted* prices are martingales.
- ➋ To *use* it we must actually find  $\mathbb{Q}$  — this is what **Girsanov's** theorem delivers (an intermezzo).
- ➌ **Apply it:** under  $\mathbb{Q}$  the stock earns  $r$ , and the option price is a discounted  $\mathbb{Q}$ -expectation. Worked examples (digital put, Philips put).
- ➍ An equivalent encoding: the **pricing kernel**.
- ➎ When is  $\mathbb{Q}$  *unique*? The **Second Fundamental Theorem** (completeness).

## Section 1

### Risk-neutral pricing

In part 1 we priced options in the one-period toy model two ways: by *replication*, and by *risk-neutral* pricing under a measure  $\mathbb{Q}$  that made  $S/B$  a martingale. There we proved the First Fundamental Theorem of asset pricing.

The same ideas carry over to continuous time — only the machinery is heavier. We now

- state the continuous-time First Fundamental Theorem;
- *find* the risk-neutral measure  $\mathbb{Q}$  (via Girsanov);
- use it to price European options.

# The First Fundamental Theorem of asset pricing

## Theorem

*Under regularity conditions: absence of arbitrage holds if and only if, for some numéraire  $N$ , there exists a probability measure*

*$\mathbb{Q} = \mathbb{Q}_N$  such that:*

- (1)  $\mathbb{Q}$  is equivalent to  $\mathbb{P}$ ;
  - i.e.  $\mathbb{Q}(A) > 0$  iff  $\mathbb{P}(A) > 0$  and  $\mathbb{Q}(A) = 0$  iff  $\mathbb{P}(A) = 0$ ;*
- (2) for any asset  $A$  in the market: the discounted price process  $A/N$  is a  $\mathbb{Q}$ -martingale, i.e.*

$$\frac{A_t}{N_t} = \mathbb{E}_{\mathbb{Q}} \left[ \frac{A_{t+h}}{N_{t+h}} \mid \mathcal{F}_t \right] = \mathbb{E}_{\mathbb{Q}} \left[ \frac{A_T}{N_T} \mid \mathcal{F}_t \right].$$

## Remarks:

- any traded asset  $A$  with  $\mathbb{P}(A_t > 0) = 1$  for all  $t$  can be used as numéraire
- typically the Money Market Account  $B$  is used as numéraire

We will only give a full proof for the Black-Scholes market. To prepare for the proof we first notice the following:

- Consider Black-Scholes market
- Use  $N = B$
- If  $V$  is self-financing portfolio then

$$\begin{aligned}d\frac{V_t}{B_t} &= \frac{1}{B_t} dV_t - r\frac{V_t}{B_t} dt = \frac{1}{B_t}(\phi_t dS_t + \psi_t dB_t) - r\frac{V_t}{B_t} dt \\ &= \dots = \phi_t d\frac{S_t}{B_t}\end{aligned}$$

Hence: if  $S/B$  is  $\mathbb{Q}$ -martingale, then  $V/B$  is  $\mathbb{Q}$ -martingale as well!



# First Fundamental Theorem of asset pricing

## Proof.

' $\Leftarrow$ ' (general proof; not only for Black-Scholes market)

- let  $V$  be the value process corresponding to a self-financing trading strategy
- suppose  $V$  satisfies  $\mathbb{P}(V_T < 0) = 0$  and  $\mathbb{P}(V_T > 0) > 0$ 
  - if  $\mathbb{P}(V_0 \leq 0) = 1$  we would have an arbitrage opportunity
- equivalence yields:  $\mathbb{Q}(V_T < 0) = 0$  and  $\mathbb{Q}(V_T > 0) > 0$
- we have (use previous slide!)

$$V_0 = N_0 \mathbb{E}_{\mathbb{Q}} \left[ \frac{V_T}{N_T} \mid \mathcal{F}_0 \right] = N_0 \mathbb{E}_{\mathbb{Q}} \left[ \frac{V_T}{N_T} \right] > 0$$

- consequently  $\mathbb{P}(V_0 > 0) > 0 \implies$  strategy does not yield arbitrage opportunity

' $\Rightarrow$ ' We do not prove this implication in general. For the Black-Scholes market you will see an 'explicit' construction. □

# Using the first fundamental theorem to price options

- (a) Construct model for basic assets (for example,  $S$  and  $B$ ).
- (b) Apply FFT: If your model does not allow for arbitrage opportunities, FFT says that you should be able to find  $\mathbb{Q} \sim \mathbb{P}$  such that  $S/B$  is  $\mathbb{Q}$ -martingale (assuming that you use  $B$  as numéraire, note that  $B/B$  then yields a martingale). Use this to determine  $\mathbb{Q}$ . We will (typically) find unique solution.
- (c) Now introduce an option (on  $S$ ) to the market. We know its payoff  $h(S_T)$  at maturity and want to determine its no-arbitrage price  $C_t$  for  $t < T$ .
- (d) Apply FFT:  $B/N$ ,  $S/N$ , and  $C/N$  should be  $\mathbb{Q}$ -martingales. From step (b) you already obtained unique  $\mathbb{Q}$ . This implies that we must set price of option  $C_t$ , for  $t \in [0, T)$ , via

$$C_t = N_t \mathbb{E}_{\mathbb{Q}} \left[ \frac{C_T}{N_T} \mid \mathcal{F}_t \right] = N_t \mathbb{E}_{\mathbb{Q}} \left[ \frac{h(S_T)}{N_T} \mid \mathcal{F}_t \right].$$

# Using the first fundamental theorem to price option

- price processes basic assets in standard Black-Scholes market given; in particular:

$$S_T = S_t \exp \left( \left( \mu - \frac{1}{2} \sigma^2 \right) (T - t) + \sigma (W_T^{\mathbb{P}} - W_t^{\mathbb{P}}) \right),$$

where  $W^{\mathbb{P}}$  is a standard  $\mathbb{P}$ -Brownian motion

- use these to solve for  $\mathbb{Q}$ 
  - how?**
- want to price European option with payoff  $C_T = h(S_T)$ :

$$C_t = N_t \mathbb{E}_{\mathbb{Q}} \left[ \frac{C_T}{N_T} \mid \mathcal{F}_t \right] = N_t \mathbb{E}_{\mathbb{Q}} \left[ \frac{h(S_T)}{N_T} \mid \mathcal{F}_t \right]$$

- in case  $N_t = B_t = \exp(rt)$ :

$$C_t = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} [C_T \mid \mathcal{F}_t] = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} [h(S_T) \mid \mathcal{F}_t]$$

- $\implies$  **we do not need to know  $\mathbb{Q}$ , but 'only' need to know the distribution of  $W^{\mathbb{P}}$ , and hence  $S_T$ , under  $\mathbb{Q}$ !**

## Theorem (Girsanov)

*Suppose the process  $W^{\mathbb{P}}$  is a standard Brownian motion under  $\mathbb{P}$  and let  $T < \infty$ . Consider a process  $(\gamma_t)_{t \geq 0}$  adapted to the filtration  $(\mathcal{F}_t)_{t \geq 0}$  generated by  $W^{\mathbb{P}}$  and satisfying appropriate integrability conditions. Define  $\mathbb{Q}$  by, for events  $A \in \mathcal{F}_T$ ,*

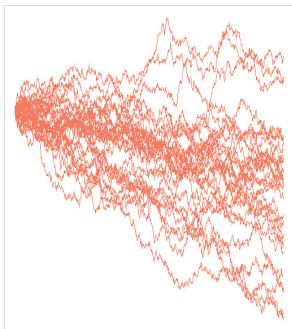
$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}} \left[ 1_A \exp \left( -\frac{1}{2} \int_0^T \gamma_t^2 dt - \int_0^T \gamma_t dW_t^{\mathbb{P}} \right) \right].$$

*Then we have:*

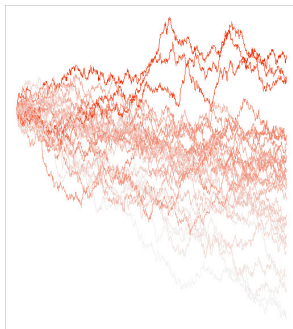
- $\mathbb{Q}$  is a probability measure equivalent to  $\mathbb{P}$ ,
- and the process  $W^{\mathbb{Q}}$  defined by

$$W_0^{\mathbb{Q}} = 0, \quad dW_t^{\mathbb{Q}} = \gamma_t dt + dW_t^{\mathbb{P}},$$

*or equivalently  $W_t^{\mathbb{Q}} = \int_0^t \gamma_u du + W_t^{\mathbb{P}}$ , is a standard Brownian motion under  $\mathbb{Q}$ .*



30 paths of a Brownian motion with negative drift



The same paths reweighted according to the Girsanov formula

- source: Wikipedia
- under  $\mathbb{Q}$  and  $\mathbb{P}$  same sample paths possible (equivalence), but with different likelihoods
- a sketch of proof is given in the appendix

- Girsanov: for

$$W_0^{\mathbb{Q}} = 0, \quad dW_t^{\mathbb{Q}} = \gamma_t dt + dW_t^{\mathbb{P}} \quad (W^{\mathbb{P}} \text{ standard BM under } \mathbb{P})$$

we can find  $\mathbb{Q}$  equivalent to  $\mathbb{P}$  such that  $W^{\mathbb{Q}}$  is standard Brownian motion under  $\mathbb{Q}$

- Let  $0 < a \neq 1$ . Suppose that we can find probability measure  $\mathbb{Q}$  such that

$$Z_0^{\mathbb{Q}} = 0, \quad dZ_t^{\mathbb{Q}} = \gamma_t dt + a dW_t^{\mathbb{P}} \quad (W^{\mathbb{P}} \text{ standard BM under } \mathbb{P}),$$

is standard Brownian motion under  $\mathbb{Q}$ . Then  $\mathbb{Q}$  is *not* equivalent to  $\mathbb{P}$ .

**Recall the First Fundamental Theorem** (stated earlier): for a numéraire  $N$ , absence of arbitrage means there is a measure  $\mathbb{Q} \sim \mathbb{P}$  under which *every* discounted price  $A/N$  is a martingale.

**Plan.** In the Black-Scholes market we now *construct* this  $\mathbb{Q}$  explicitly — using Girsanov — and then read off option prices as discounted  $\mathbb{Q}$ -expectations.

(We use the money-market account  $B$  as numéraire; any asset that stays strictly positive would also do.)

# Using FFT and Girsanov: setup

- setup: standard Black-Scholes market ( $W = W^{\mathbb{P}}$  is standard Brownian motion):

$$dB_t = rB_t dt, \quad dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}, \quad S_0 > 0, \quad B_0 = 1$$

- use  $B$  as numéraire
- Easy exercise (Itô) to show that

$$d\frac{S_t}{B_t} = (\mu - r)\frac{S_t}{B_t} dt + \sigma\frac{S_t}{B_t} dW_t^{\mathbb{P}}$$

- FFT  $\implies$  we need to find  $\mathbb{Q}$  such that  $S_t/B_t$  is  $\mathbb{Q}$ -martingale
  - note that  $B/B$  is automatically a martingale



# Using FFT and Girsanov: finding $\mathbb{Q}$

$$d \frac{S_t}{B_t} = (\mu - r) \frac{S_t}{B_t} dt + \sigma \frac{S_t}{B_t} dW_t^{\mathbb{P}}$$

- for

$$dW_t^{\mathbb{Q}} = \gamma_t dt + dW_t^{\mathbb{P}}$$

we have  $dW_t^{\mathbb{P}} = -\gamma_t dt + dW_t^{\mathbb{Q}}$  which yields

$$d \frac{S_t}{B_t} = ((\mu - r) - \sigma \gamma_t) \frac{S_t}{B_t} dt + \sigma \frac{S_t}{B_t} dW_t^{\mathbb{Q}}$$

- application of Girsanov's theorem with

$$\gamma_t = \frac{\mu - r}{\sigma} \quad \text{market price of risk}$$

yields  $\mathbb{Q} \sim \mathbb{P}$  such that  $W^{\mathbb{Q}}$  is a  $\mathbb{Q}$ -standard Brownian motion

- we have

$$d \frac{S_t}{B_t} = \sigma \frac{S_t}{B_t} dW_t^{\mathbb{Q}}$$

so  $S/B$  is martingale under  $\mathbb{Q}$  (not under  $\mathbb{P}$ !)

# From the binomial model to the market price of risk

In part 1 the risk-neutral probability  $q$  tilted the real-world  $p$  just enough to make  $S/B$  a martingale. In continuous time Girsanov does the same job continuously, through the **market price of risk**

$$\gamma = \frac{\mu - r}{\sigma}.$$

- $\mu - r$  is the stock's *excess return* over the riskless rate; dividing by  $\sigma$  gives excess return *per unit of risk* (a Sharpe ratio);
- Girsanov removes exactly this drift,  $dW_t^{\mathbb{Q}} = \gamma dt + dW_t^{\mathbb{P}}$ , turning the stock's drift from  $\mu$  into  $r$ ;
- discrete analogue (part 1):  $q = \frac{B_1/B_0 - s_d/S_0}{s_u/S_0 - s_d/S_0}$  is the same tilt, one period at a time.

The larger the risk premium  $\mu - r$  (or the smaller the risk  $\sigma$ ), the larger the adjustment  $\gamma$ .

## Remark:

Under  $\mathbb{Q}$  we have

$$d \frac{S_t}{B_t} = \sigma \frac{S_t}{B_t} dW_t^{\mathbb{Q}}$$

$\implies$  (use Itô)

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

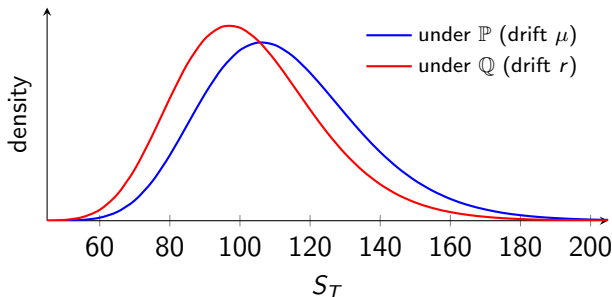
- under  $\mathbb{P}$  we have

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}$$

- under  $\mathbb{Q}$  stock has same drift as bond, but with risk  $\implies$   
**risk-neutral (world)**

# The real-world and risk-neutral laws of $S_T$

The change of measure  $\mathbb{P} \rightarrow \mathbb{Q}$  shifts the stock's drift from  $\mu$  to  $r$  but leaves  $\sigma$  unchanged. So  $S_T$  is lognormal under both, with the *same* spread but a different centre:



Here  $S_0 = 100$ ,  $\sigma = 20\%$ ,  $T = 1$ ,  $\mu = 12\%$ ,  $r = 3\%$ ; then  $\mathbb{E}_{\mathbb{P}}[S_T] = 112.8$  but  $\mathbb{E}_{\mathbb{Q}}[S_T] = 103.0$ . Same outcomes (equivalent measures!), different likelihoods.

## Worked example: under $\mathbb{Q}$ the stock earns the riskless rate

Under  $\mathbb{Q}$ ,  $S_T = S_t e^{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})}$  with  $W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}} \sim N(0, T-t)$ . Using  $\mathbb{E}[e^{\sigma Z'}] = e^{\frac{1}{2}\sigma^2(T-t)}$  for  $Z' \sim N(0, T-t)$ :

$$\mathbb{E}_{\mathbb{Q}}[S_T \mid \mathcal{F}_t] = S_t e^{(r - \frac{1}{2}\sigma^2)(T-t)} e^{\frac{1}{2}\sigma^2(T-t)} = S_t e^{r(T-t)}.$$

Equivalently  $\mathbb{E}_{\mathbb{Q}}[S_T/B_T \mid \mathcal{F}_t] = S_t/B_t$ :  $S/B$  is a  $\mathbb{Q}$ -martingale, as it must be. The same computation under  $\mathbb{P}$  gives  $\mathbb{E}_{\mathbb{P}}[S_T \mid \mathcal{F}_t] = S_t e^{\mu(T-t)}$  — the stock earns its true drift  $\mu > r$ , a *risk premium*.

This is the continuous-time version of the binomial identity

$$\frac{1}{B_1} \mathbb{E}_{\mathbb{Q}}[S_1] = S_0 \text{ from part 1.}$$

# Using FFT and Girsanov: the pricing formula

- the SDE

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

is GBM

- so, under  $\mathbb{Q}$ ,

$$S_T = S_t e^{(r-0.5\sigma^2)(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})}$$

- option with payoff  $h(S_T)$
- FFT yields

$$\begin{aligned} C_t &= e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left[ h \left( S_t e^{(r-0.5\sigma^2)(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})} \right) \mid \mathcal{F}_t \right] \\ &= e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}} \left[ h \left( S_t e^{(r-0.5\sigma^2)(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})} \right) \right] \quad (S_t \text{ is constant}) \end{aligned}$$

- using  $W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}} \sim \sqrt{T-t}Z$  (under  $\mathbb{Q}$ ) with  $Z \sim N(0, 1)$ :

$$C_t = e^{-r(T-t)} \int_{-\infty}^{\infty} h \left( S_t e^{(r-0.5\sigma^2)(T-t) + \sigma\sqrt{T-t}z} \right) \phi(z) dz$$

## Example - digital put

- digital European put:  $h(S_T) = 1\{S_T \leq K\}$ ; what is the price at  $t = 0$ ?
- we have to calculate

$$C_0 = \exp(-rT) \mathbb{E}_{\mathbb{Q}} [h(S_T)] = \exp(-rT) \mathbb{Q}(S_T \leq K)$$

- using

$$\begin{aligned} S_T &= S_0 \exp\left((r - 0.5\sigma^2)T + \sigma W_T^{\mathbb{Q}}\right) \leq K \\ &\Leftrightarrow \sigma W_T^{\mathbb{Q}} \leq \log \frac{K}{S_0} - (r - 0.5\sigma^2)T \end{aligned}$$

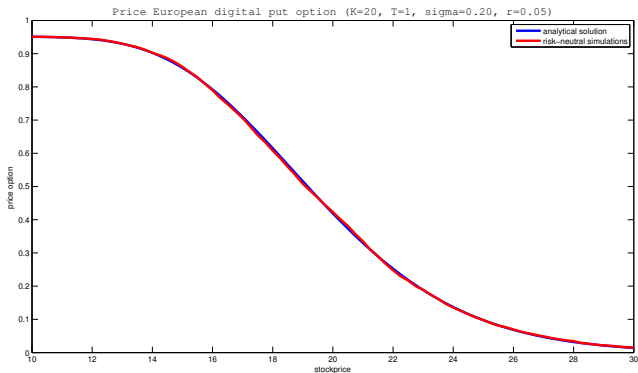
we obtain

$$\begin{aligned} C_0 &= \exp(-rT) \mathbb{Q}\left(\sigma W_T^{\mathbb{Q}} \leq \log \frac{K}{S_0} - (r - 0.5\sigma^2)T\right) \\ &= \exp(-rT) \Phi\left(\frac{\log \frac{K}{S_0} - (r - 0.5\sigma^2)T}{\sigma\sqrt{T}}\right) \end{aligned}$$

This is exactly the price the part-2 PDE approach gives for the digital put — the two methods agree.

## Example digital put

The blue line plots  $C_0$  as function of  $S_0$  for  $K = 20$ ,  $T = 1$ ,  $\sigma = 20\%$ , and  $r = 5\%$ .



Later on we will discuss how we can use Monte Carlo techniques to obtain a numerical approximation. The red line is such an approximation based on 5,000 simulations.



## Quiz

In the Black-Scholes market with  $\mu = 10\%$ ,  $r = 2\%$ ,  $\sigma = 25\%$ :

- 1 What is the market price of risk  $\gamma$ ?
- 2 Write the SDE for  $S$  under the risk-neutral measure  $\mathbb{Q}$ .
- 3 What is the distribution of  $\log S_T$  under  $\mathbb{Q}$  (given  $S_0$ )?
- 4 Is  $\mathbb{E}_{\mathbb{Q}}[S_T]$  larger or smaller than  $\mathbb{E}_{\mathbb{P}}[S_T]$ ?

# Example: put on Philips



historical data yields estimated  $\sigma = 0.29$

# Example: put on Philips

| DECEMBER 2021 PRICES - 17/07/20 |        |      |       |       |   |        |   |       |      |        |
|---------------------------------|--------|------|-------|-------|---|--------|---|-------|------|--------|
|                                 | SETTL. | LAST | BID   | ASK   |   | STRIKE |   | BID   | LAST | SETTL. |
| +                               | 18.02  | -    | 17.70 | 18.00 | C | 25.00  | P | 0.46  | -    | 0.52   |
| +                               | 13.63  | -    | 13.30 | 13.60 | C | 30.00  | P | 1.12  | -    | 1.17   |
| +                               | 9.75   | -    | 9.45  | 9.75  | C | 35.00  | P | 2.29  | -    | 2.29   |
| +                               | 6.48   | -    | 6.20  | 6.50  | C | 40.00  | P | 4.07  | -    | 4.06   |
| +                               | 3.98   | -    | 3.81  | 4.00  | C | 45.00  | P | 6.60  | -    | 6.60   |
| +                               | 2.24   | -    | 2.14  | 2.29  | C | 50.00  | P | 9.95  | -    | 9.93   |
| +                               | 0.60   | -    | 0.52  | 0.67  | C | 60.00  | P | 18.40 | -    | 18.37  |

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- $T = 17.5/12$ ,  $r = 0.01$ ,  $K = 45$ ,  $\sigma = 0.29$  and  $S_0 = 43.145$  yields 6.70 as B-S price for the put, where  $\sigma = 0.29$  is result from estimation on historical data on  $S$ .

## Food for thought

The risk-neutral measure  $\mathbb{Q}$  is *not* the real-world measure: it never describes the actual probabilities of price moves, and under  $\mathbb{Q}$  every asset earns only  $r$ .

- In what sense, then, is  $\mathbb{Q}$  “fictional but useful”?
- Two investors who disagree about the drift  $\mu$  (hence about  $\mathbb{P}$ ) still agree on the option price. Why does  $\mu$  drop out?
- If a second, *untraded* source of risk is added,  $\mathbb{Q}$  is no longer unique (Second FTAP). What breaks in the replication argument?

## Section 2

### Pricing Kernels

# Motivation

- suppose we work in standard Black-Scholes market with  $B$  as numéraire
- Girsanov yields probability measure  $\mathbb{Q}$  defined by  $\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}} 1_A Z_T$  where

$$Z_t = \exp \left( -t(\mu - r)^2 / (2\sigma^2) - ((\mu - r)/\sigma) W_t^{\mathbb{P}} \right)$$

- we know  $\mathbb{E}_{\mathbb{Q}}[X] = \mathbb{E}_{\mathbb{P}}[XZ_T]$
- it can be shown that  $\mathbb{E}_{\mathbb{Q}}[X|\mathcal{F}_t] = \mathbb{E}_{\mathbb{P}}[X(Z_T/Z_t)|\mathcal{F}_t]$
- we thus have

$$A_t = \mathbb{E}_{\mathbb{Q}} \left[ \frac{N_t}{N_T} A_T \mid \mathcal{F}_t \right] = \mathbb{E}_{\mathbb{P}} \left[ \frac{N_t}{N_T} \frac{Z_T}{Z_t} A_T \mid \mathcal{F}_t \right]$$

which yields, with  $K_t = Z_t/N_t$ ,

$$K_t A_t = \mathbb{E}_{\mathbb{P}}[K_T A_T \mid \mathcal{F}_t],$$

i.e.  $KA$  is a  $\mathbb{P}$ -martingale

Here  $K = Z/N$  is the discounted likelihood ratio  $\frac{1}{N} \frac{d\mathbb{Q}}{d\mathbb{P}}$  — the continuous-time version of the kernel  $m = \frac{B_0}{B_1} \frac{d\mathbb{Q}}{d\mathbb{P}}$  from part 1.

# The first fundamental theorem of asset pricing (pricing kernel version)

We proved, for the special case of the Black-Scholes market, the following theorem.

## Theorem

*Absence of arbitrage holds if and only if there exists a process  $K$ , with  $K_t > 0$ , such that for any asset with price  $A_t$  the process  $K_t A_t$  is a  $\mathbb{P}$ -martingale*

## Application:

- determine  $K$  (using your models for the basic assets)
- (no arbitrage) price of a derivative, with payoff  $C_T$ , based on these assets is given by

$$C_t = \frac{1}{K_t} \mathbb{E}_{\mathbb{P}}[K_T C_T \mid \mathcal{F}_t]$$

# The pricing kernel for the standard Black-Scholes world

- try  $K$  of the form  $dK_t = \mu_t^K dt + \sigma_t^K dW_t^{\mathbb{P}}$
- Itô's product rule yields

$$\begin{aligned}dK_t B_t &= K_t dB_t + B_t dK_t + d[K, B]_t \\&= (rK_t B_t + B_t \mu_t^K) dt + \dots dW_t^{\mathbb{P}}\end{aligned}$$

$$K_t B_t \text{ martingale} \implies \text{no drift} \implies \mu_t^K = -rK_t$$

- Itô's product rule yields

$$\begin{aligned}dK_t S_t &= K_t dS_t + S_t dK_t + d[S, K]_t \\&= (K_t \mu S_t + S_t \mu_t^K + \sigma \sigma_t^K S_t) dt + \dots dW_t^{\mathbb{P}}\end{aligned}$$

$$K_t S_t \text{ martingale} \implies \text{no drift} \implies \sigma_t^K = -K_t(\mu - r)/\sigma$$

So

$$K_0 = 1, \quad dK_t = -rK_t dt - \frac{\mu - r}{\sigma} K_t dW_t^{\mathbb{P}}$$

determines the pricing kernel process in the B-S model.



# The pricing kernel for the standard Black-Scholes world

$$K_0 = 1, \quad dK_t = -rK_t dt - \frac{\mu - r}{\sigma} K_t dW_t^{\mathbb{P}}$$

has as solution

$$K_t = \exp \left\{ \left( -r - 0.5 \left( \frac{\mu - r}{\sigma} \right)^2 \right) t - \frac{\mu - r}{\sigma} W_t^{\mathbb{P}} \right\}$$

Fair price of a derivative, with payoff  $C_T = F(S_T)$ , based on these assets is given by

$$C_0 = \frac{1}{K_0} \mathbb{E}_{\mathbb{P}}[K_T C_T]$$

- $K_T C_T = g(W_T) f(S_T) = g(W_T) \tilde{f}(W_T) = h(W_T)$ , so price determined by expectation of functional of  $N(0, T)$  variable
- price seems to depend on  $\mu$ , however it does not!

## Remark

- you can always set  $K_0 = 1$
- which yields

$$C_0 = \mathbb{E}_{\mathbb{P}}[K_T C_T]$$

- suppose  $C_T = 1_{\{\omega \in E\}}$ , i.e. the pay-off is 1 euro in case event  $E$  occurs and 0 otherwise
- then

$$C_0 = \mathbb{E}_{\mathbb{P}}[1_E K_T],$$

so  $K_T(\omega)$  kind of measures how expensive it is to get a pay-off in state  $\omega$

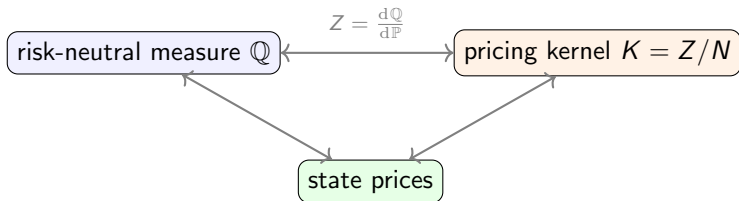
- For the Black-Scholes market we have (assume  $\mu > r$ )

$$K_T = \exp \left\{ \left( -r - 0.5 \left( \frac{\mu - r}{\sigma} \right)^2 \right) T - \frac{\mu - r}{\sigma} W_T \right\},$$

showing that  $K_T$  takes large values for large, negative values of  $W_T$ . And such state-of-the-world correspond to small values of  $S_T(\omega)$

# One object, three encodings

The risk-neutral measure, the pricing kernel, and state prices are three views of the *same* pricing rule.



The same price, three ways:

$$C_t = N_t \mathbb{E}_{\mathbb{Q}} \left[ \frac{C_T}{N_T} \mid \mathcal{F}_t \right] = \frac{1}{K_t} \mathbb{E}_{\mathbb{P}} [K_T C_T \mid \mathcal{F}_t] = \int C_T d\pi.$$

In the part-1 toy model these were the measure  $\mathbb{Q}$ , the kernel  $m$ , and the Arrow-Debreu prices  $\pi_u, \pi_d$ .

## Section 3

### Second Fundamental Theorem of asset pricing

In our discussion we have assumed that it is possible to find a self-financing portfolio for every option pay-off  $C_T$ . This property is referred to as 'completeness'.

## Definition

A financial market is said to be **complete** if it is possible to find, for each  $T > 0$ , and all random variables  $C_T$  that are  $\mathcal{F}_T$ -measurable, a self-financing portfolio  $V$  (satisfying regularity conditions) that satisfies  $V_T = C_T$  (a.s.)

# Second Fundamental Theorem of asset pricing

## Theorem

Under regularity conditions we have: a financial market is complete if and only if there is a *unique* measure  $\mathbb{Q} \sim \mathbb{P}$  that makes all deflated asset prices  $A/N$   $\mathbb{Q}$ -martingales (where  $N$  is the chosen numéraire).

## Examples

- For our Black-Scholes market  $\mathbb{Q}$  was unique, hence the market is complete.
- Consider the Black-Scholes market. Now we also add an additional risk factor to the market: an independent Brownian motion. This has impact on  $\mathcal{F}_T$ ! And now claims could depend on both Brownian motions. No additional assets are available. This leads to an incomplete market. (Why is  $\mathbb{Q}$  not unique?)

**Remark** In the MSc in QFAS we discuss valuation (and hedging) in incomplete markets (which you often encounter in case there is financial risk as well as actuarial risk).

# Three methods, one price

For the European digital put (payoff  $1\{S_T \leq K\}$ ) all three methods give the *same* price:

| method                        | price $C_0$         |
|-------------------------------|---------------------|
| Black-Scholes PDE (part 2)    | $e^{-rT} \Phi(d_-)$ |
| risk-neutral pricing (part 3) | $e^{-rT} \Phi(d_-)$ |
| pricing kernel (part 3)       | $e^{-rT} \Phi(d_-)$ |

with  $d_- = \frac{\log(K/S_0) - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ . The PDE, the risk-neutral measure, and the pricing kernel are three routes to the *same* no-arbitrage price — exactly as in the binomial toy model of part 1.

## Section 4

### Appendix



## Appendix: sketch proof of Girsanov (1/2)

- given the integrability conditions, it is easy to show that  $\mathbb{Q}$  is probability measure equivalent to  $\mathbb{P}$
- to prove that  $W^{\mathbb{Q}}$  is a standard Brownian motion under  $\mathbb{Q}$  we need to show, for any  $0 = t_0 < t_1 < \dots < t_n \leq T$ , that

$$(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}, \dots, W_{t_n}^{\mathbb{Q}} - W_{t_{n-1}}^{\mathbb{Q}}) \sim N(0_n, \Sigma)$$

where  $\Sigma_{ij} = 0$  for  $i \neq j$  and  $\Sigma_{ii} = t_i - t_{i-1}$

- we only consider case in which  $u \mapsto \gamma_u$  is deterministic and  $n = 2$  with  $t_2 = T$
- it is sufficient to show (MGF determines the probability distribution)

$$\mathbb{E}_{\mathbb{Q}} \left[ e^{a_1(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}) + a_2(W_{t_2}^{\mathbb{Q}} - W_{t_1}^{\mathbb{Q}})} \right] = e^{\frac{1}{2}(a_1^2(t_1 - t_0) + a_2^2(t_2 - t_1))}$$

## Appendix: sketch proof of Girsanov (2/2)

We have

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}} \left[ e^{a_1(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}) + a_2(W_{t_2}^{\mathbb{Q}} - W_{t_1}^{\mathbb{Q}})} \right] &= \mathbb{E}_{\mathbb{P}} \left[ \frac{d\mathbb{Q}}{d\mathbb{P}} e^{a_1(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}) + a_2(W_{t_2}^{\mathbb{Q}} - W_{t_1}^{\mathbb{Q}})} \right] \\&= \mathbb{E}_{\mathbb{P}} \left[ e^{a_1(W_{t_1}^{\mathbb{Q}} - W_{t_0}^{\mathbb{Q}}) + a_2(W_{t_2}^{\mathbb{Q}} - W_{t_1}^{\mathbb{Q}})} e^{-\frac{1}{2} \int_0^T \gamma_u^2 du - \int_0^T \gamma_u dW_u^{\mathbb{P}}} \right] \\&= e^{a_1 \int_{t_0}^{t_1} \gamma_u du + a_2 \int_{t_1}^{t_2} \gamma_u du - \frac{1}{2} \int_{t_0}^{t_2} \gamma_u^2 du} \\&\quad \times \mathbb{E}_{\mathbb{P}} \left[ e^{-\int_{t_0}^{t_1} (\gamma_u - a_1) dW_u^{\mathbb{P}} - \int_{t_1}^{t_2} (\gamma_u - a_2) dW_u^{\mathbb{P}}} \right]\end{aligned}$$

Exploiting independent increments and

$\int_a^b f_u dW_u^{\mathbb{P}} \sim N(0, \int_a^b f_u^2 du)$  for deterministic functions  $f$ :

$$\begin{aligned}\mathbb{E}_{\mathbb{P}} \left[ e^{-\int_{t_0}^{t_1} (\gamma_u - a_1) dW_u^{\mathbb{P}} - \int_{t_1}^{t_2} (\gamma_u - a_2) dW_u^{\mathbb{P}}} \right] \\&= \mathbb{E}_{\mathbb{P}} \left[ e^{-\int_{t_0}^{t_1} (\gamma_u - a_1) dW_u^{\mathbb{P}}} \right] \mathbb{E}_{\mathbb{P}} \left[ e^{-\int_{t_1}^{t_2} (\gamma_u - a_2) dW_u^{\mathbb{P}}} \right] \\&= e^{\frac{1}{2} \int_{t_0}^{t_1} (\gamma_u - a_1)^2 du} e^{\frac{1}{2} \int_{t_1}^{t_2} (\gamma_u - a_2)^2 du} = e^{\frac{1}{2} \int_{t_0}^{t_2} \gamma_u^2 du - a_1 \int_{t_0}^{t_1} \gamma_u du - a_2 \int_{t_1}^{t_2} \gamma_u du} \\&\quad \times e^{\frac{1}{2} (a_1^2(t_1 - t_0) + a_2^2(t_2 - t_1))} \text{ which yields the result}\end{aligned}$$